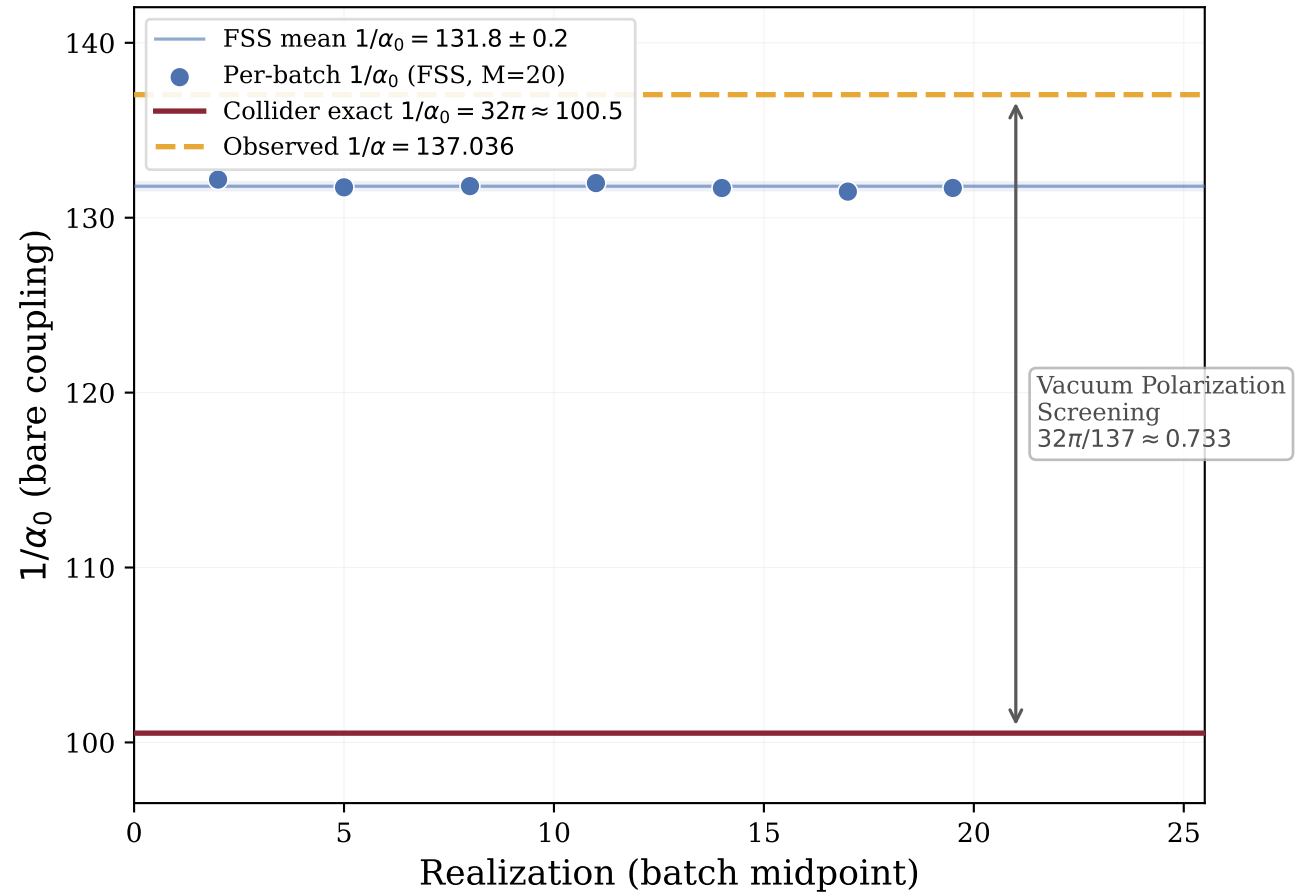




Per-batch bare coupling $1/\alpha_0$ from the FSS estimator $Q = \Sigma|\Phi|^2/\Sigma N^2$ on the ensemble topology snapshots. The topological collider (M13) measures $Q_{\text{topo}} = 1/4$ exactly, giving $\alpha_0 = 1/(32\pi) \approx 1/100.5$ at the Planck scale. Vacuum polarization screens the bare coupling to the observed $\alpha \approx 1/137$; the screening ratio is $32\pi/137 \approx 0.733$. The simulation naturally reproduces the three-generation mass hierarchy, maximal parity violation for leptons, and a Dark Matter fraction of 63%.